

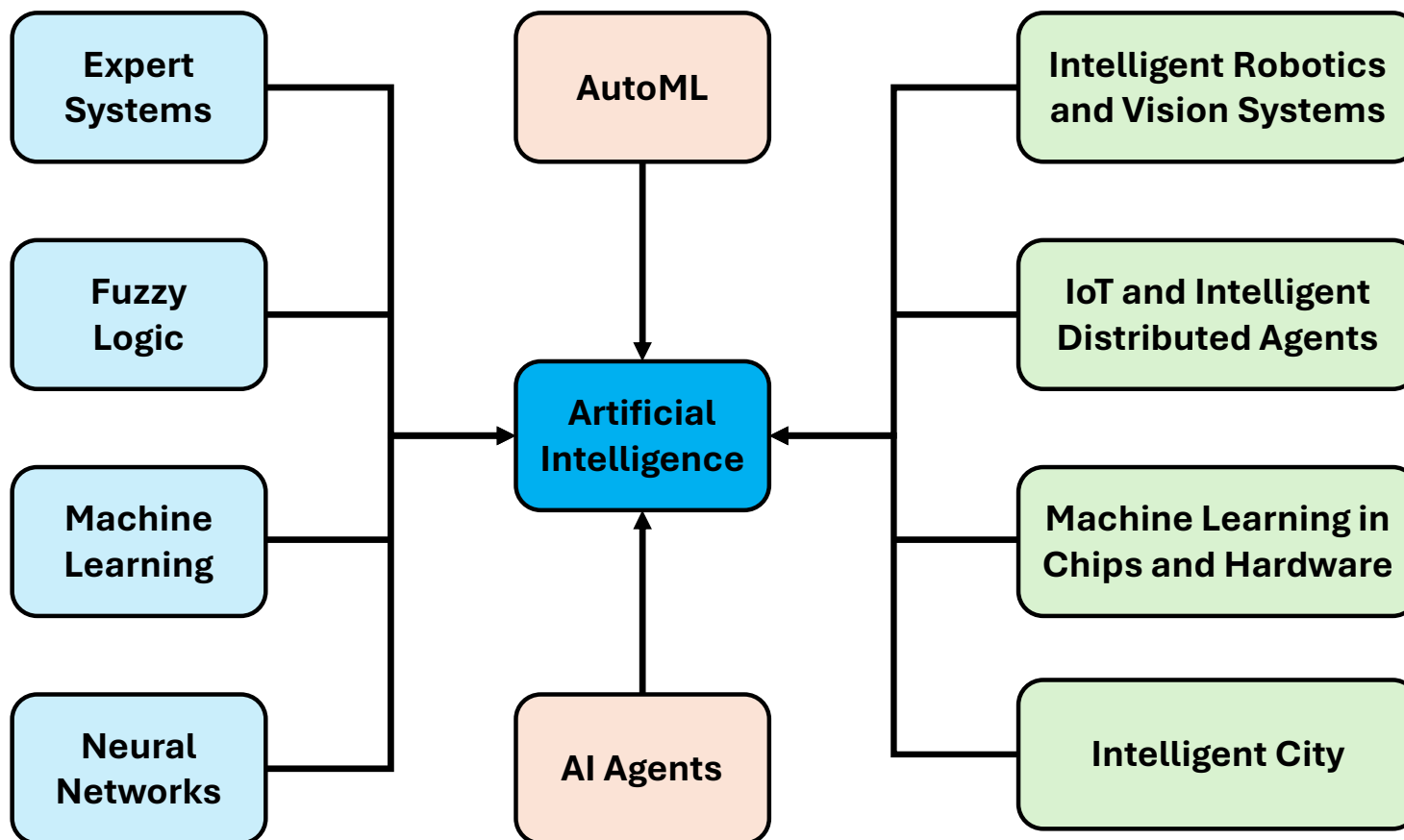


Team Tiger

ITAI 1370 – Artificial Intelligence

Lecture 2: AI Glossary Kickstart

Version 7.0



AI Agent

An AI Agent is a software system that can perceive its environment, make decisions, and take actions to achieve a goal. Unlike a traditional program that only follows fixed instructions, an AI agent can adapt its behavior based on new information.

Four Basic Components of an AI Agent

Perception: Perceive the Environment

Example Camera, Microphone, and Sensors

Reasoning: Analyze Information

Example: Machine Learning Models

Decision Making: Select the Best Action

Example: Choose the Fastest Route

Action: Perform Task

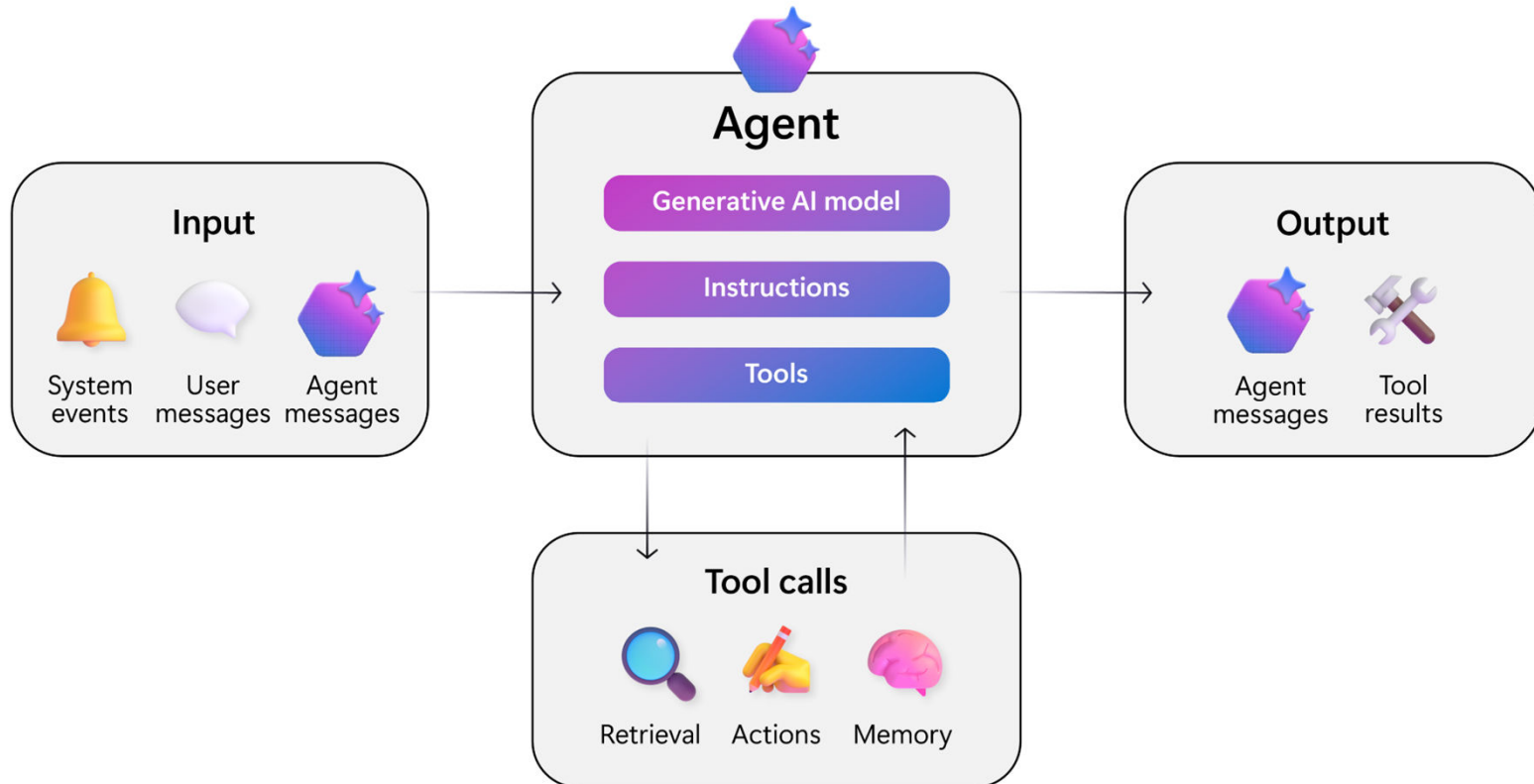
Example: Drive, Answer a Question, and Send an email

Real World Example: Self-Driving Car

The Smart Car detects roads, pedestrians, plans a route, and control steering, braking, and acceleration.

Goal: Safely Transport Passengers.

AI Agent

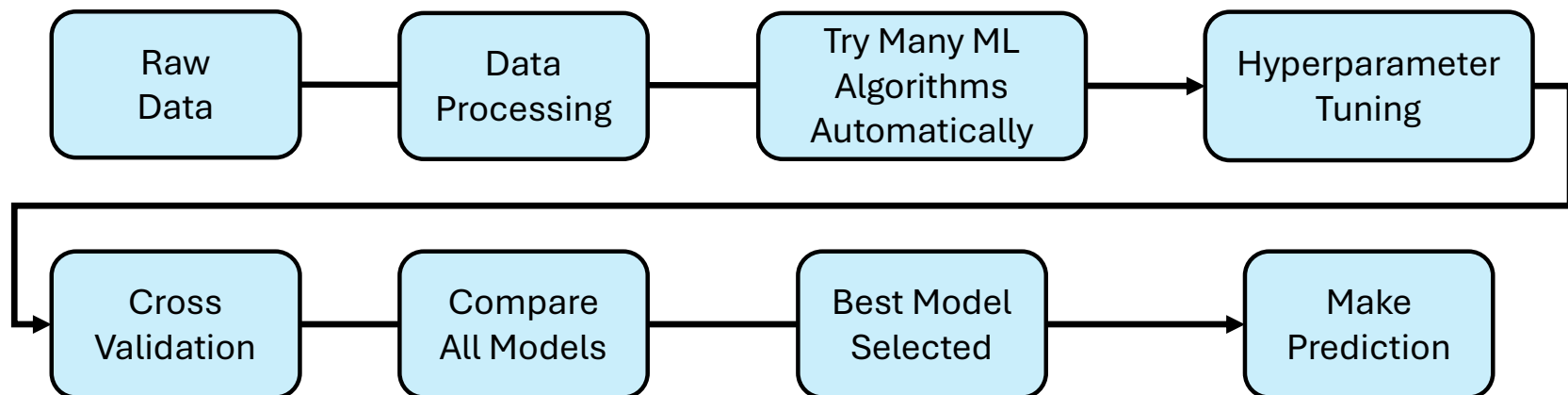


AutoML

AutoML (Automated Machine Learning)

AutoML automates data preprocessing, feature engineering, model selection, hyperparameter tuning, model evaluation, and deployment assistance.

How AutoML Works

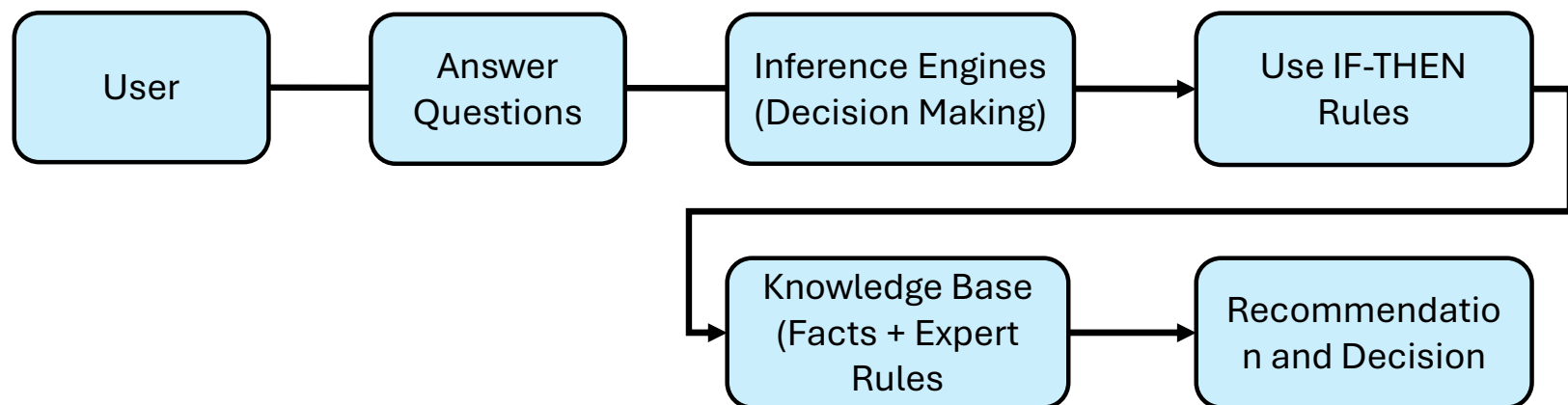


Expert System

An Expert System is a type of Artificial Intelligence (AI) that imitates the decision-making process of a human expert in a specific field.

Classical expert systems rely on expert-defined IF–THEN rules rather than learning patterns from training data.

How Expert System Works



Example: Loan Approval Rules:

IF income > \$50,000 AND credit score > 700 THEN
approve loan

Fuzzy Logic

Traditional digital logic uses only two values (0 or 1), while fuzzy logic allows degrees of truth between 0 and 1.

We must convert temperature from real value to fuzzy term like cold, warm, and hot and low, normal, and high for humidity and this process is called Fuzzification.

The second phase is Rule Evaluation, where IF–THEN rules are applied.:

If Temperature is Warm Then

Fan = Medium

End If

If Temperature is Cold Then

Fan = Off

End If

The system might combine multiple rules if they apply.

The last process of Fuzzy Logic is Defuzzification that converts the fuzzy result into a crisp numerical output. .

Example

Fan Speed

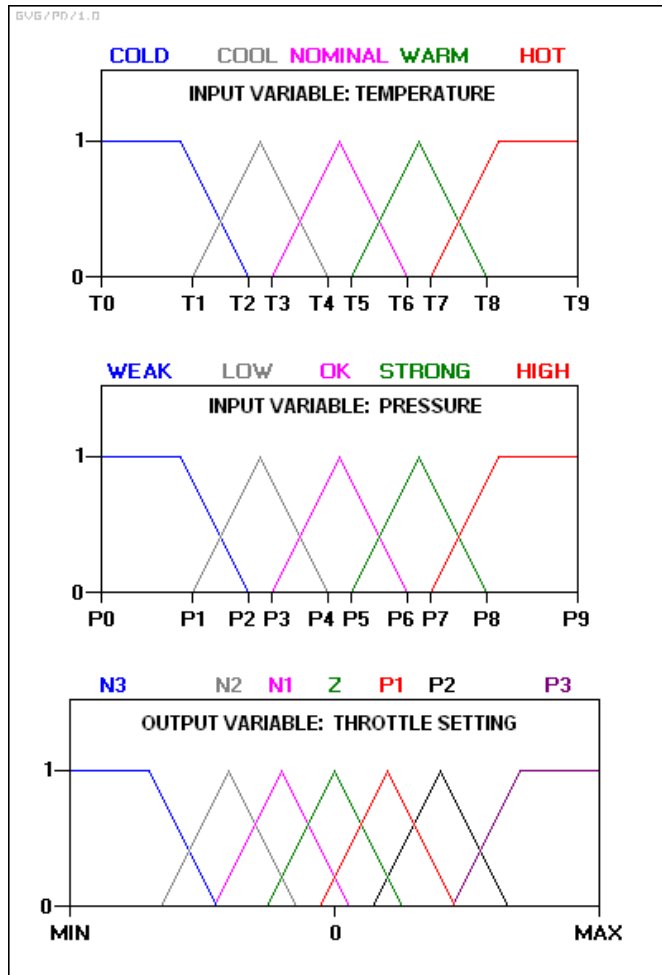
0 = Off

25 = Low

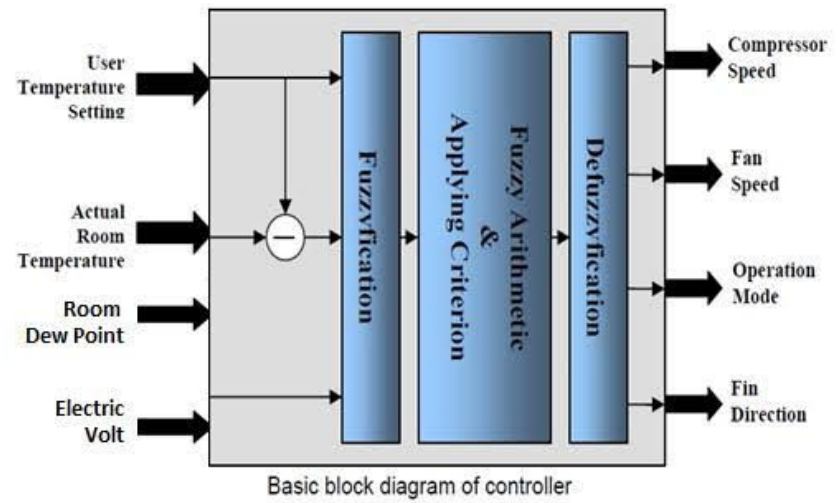
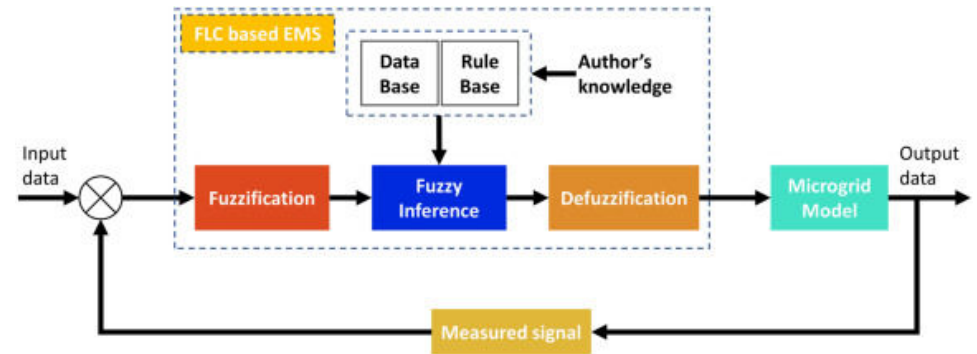
50 = Medium

75 = High

100 = Maximum



Fuzzy Logic



Intelligent City

The Core Pillars of an Intelligent City

IoT & Sensor Networks: Billions of internet-connected sensors embedded in pavement, water pipes, and buildings constantly feed raw data back to centralized platforms.

Intelligent Transit: Traffic lights that adjust dynamically based on real-time vehicle flow, smart parking apps that guide drivers to empty spots, and autonomous public transit fleets.

Predictive Infrastructure: Power grids that shift energy distribution automatically based on usage patterns, and water systems that pinpoint leak locations using acoustic sensors.

Data-Driven Decision Making: City planners use machine learning algorithms to run simulations on urban growth, disaster response, and public health trends

Smart Water: Leak detection, Smart water meters, and Flood monitoring

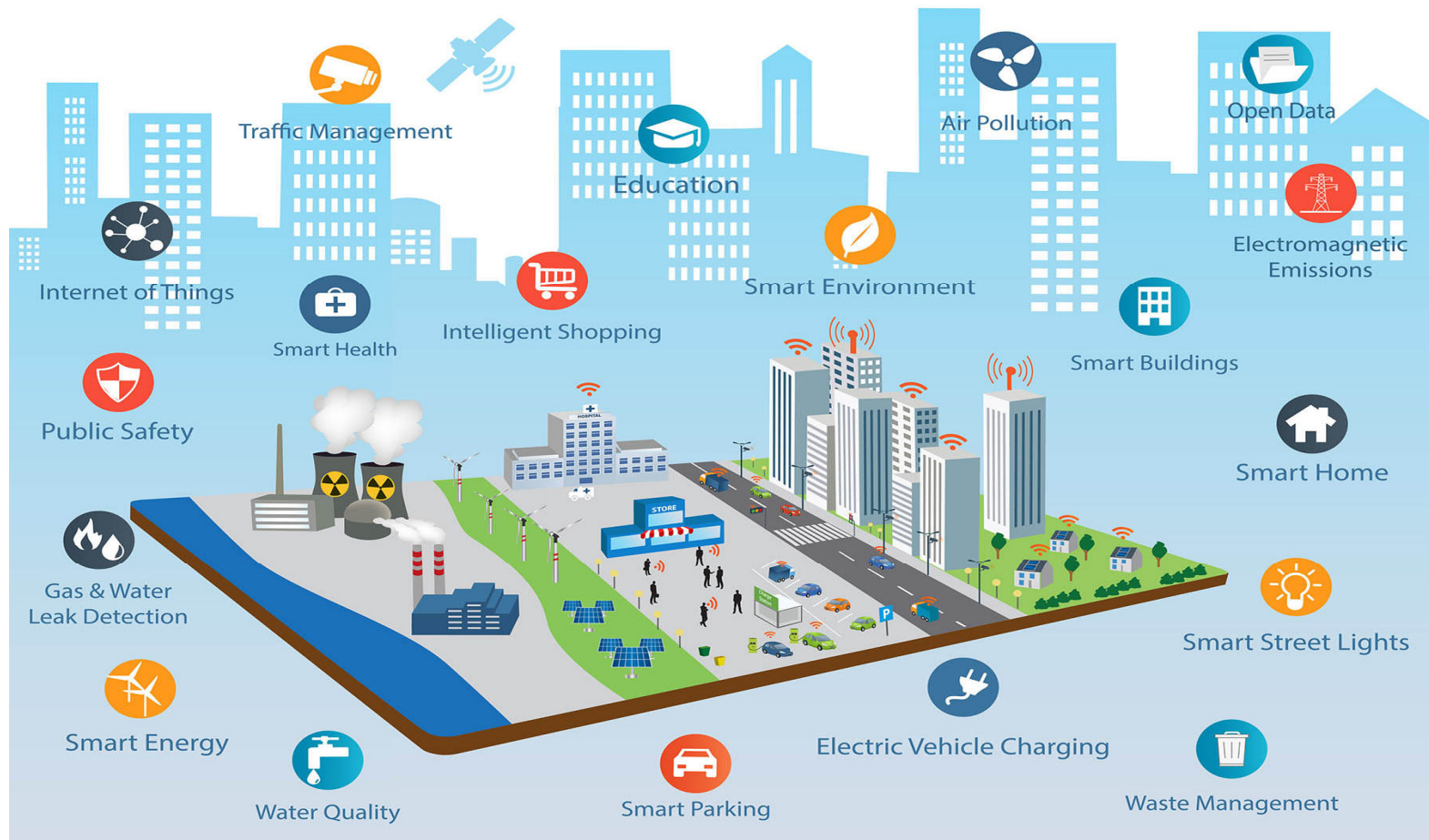
Environmental Monitoring: Air quality sensors, noise monitoring, weather stations, and pollution tracking., Noise monitoring , and Weather stations Pollution tracking

Smart Energy Smart electrical grid , that dim when no one is nearby Solar-powered infrastructure

Smart Transportation: Intelligent traffic lights , Real-time bus tracking, Smart parking, and Electric vehicle charging stations

Smart healthcare, Digital government, and Public safety

Intelligent City



Intelligent Robotic and Vision System

Main Components

Robot: The robot is the mechanical device that performs physical tasks.

Example: Industrial robot arm, Mobile robot, Drone, and Humanoid robot.

Vision System: The vision system acts as the robot's "eyes."

Example: Industrial robot arm, Mobile robot, Drone, and Humanoid robot.

Computer Vision: Computer vision processes the images and extracts useful information.

Example: Object detection, Face recognition, Barcode reading, Defect detection, Measuring object dimensions, and Motion tracking

Artificial Intelligence: AI acts as the robot's "brain."

Example: Object detection, Face recognition, Barcode reading, Defect detection, Measuring object dimensions, and Motion tracking

Controller: The controller coordinates all system components.

Example: Industrial robot arm, Mobile robot, Drone, and Humanoid robot.

Sensors: Additional sensors provide more information about the environment.

Example: Laser sensors (LiDAR), Ultrasonic sensors, Force sensors, Temperature sensors, Proximity sensors, and GPS (for outdoor robots)

Intelligent Robotic and Vision System

Applications

Manufacturing: Assembly, Welding, Painting, Quality inspection, and Packaging

Healthcare: Surgical robots, Medical image analysis, Rehabilitation robots, and Pharmacy automation

Agriculture: Crop monitoring , Fruit harvesting, and Weed detection, and Precision spraying

Transportation: Self-driving vehicles, Delivery robots, and Warehouse automation

Security: Face recognition, Surveillance, and Intruder detection

Service Industry: Hotel robots, Restaurant robots, Cleaning robots robots, and Customer assistance

Intelligent Robotic and Vision System



IoT and Intelligent Distributed Agent

IoT (Internet of Things) and **Intelligent Distributed Agents** are two important technologies in modern Artificial Intelligence and distributed computing. Together, they enable smart environments such as smart homes, smart factories, autonomous vehicles, healthcare systems, and intelligent cities.

What is IoT (Internet of Things)?

The **Internet of Things (IoT)** is a network of physical devices connected to the Internet that can: Sense the environment, Collect data, Exchange information, Make decisions, and Perform actions automatically

These devices contain: Sensors, Actuators, Processors (microcontrollers), Communication modules (Wi-Fi, Bluetooth, Zigbee, 5G, and LoRa WAN)

What is a Distributed Agent?

A **Distributed Agent** is one of many intelligent agents working together across different computers or devices. Instead of one central controller: Each agent can make local decisions independently. , Agents communicate with one another, They cooperate to solve problems, and The system continues working even if one agent fails.

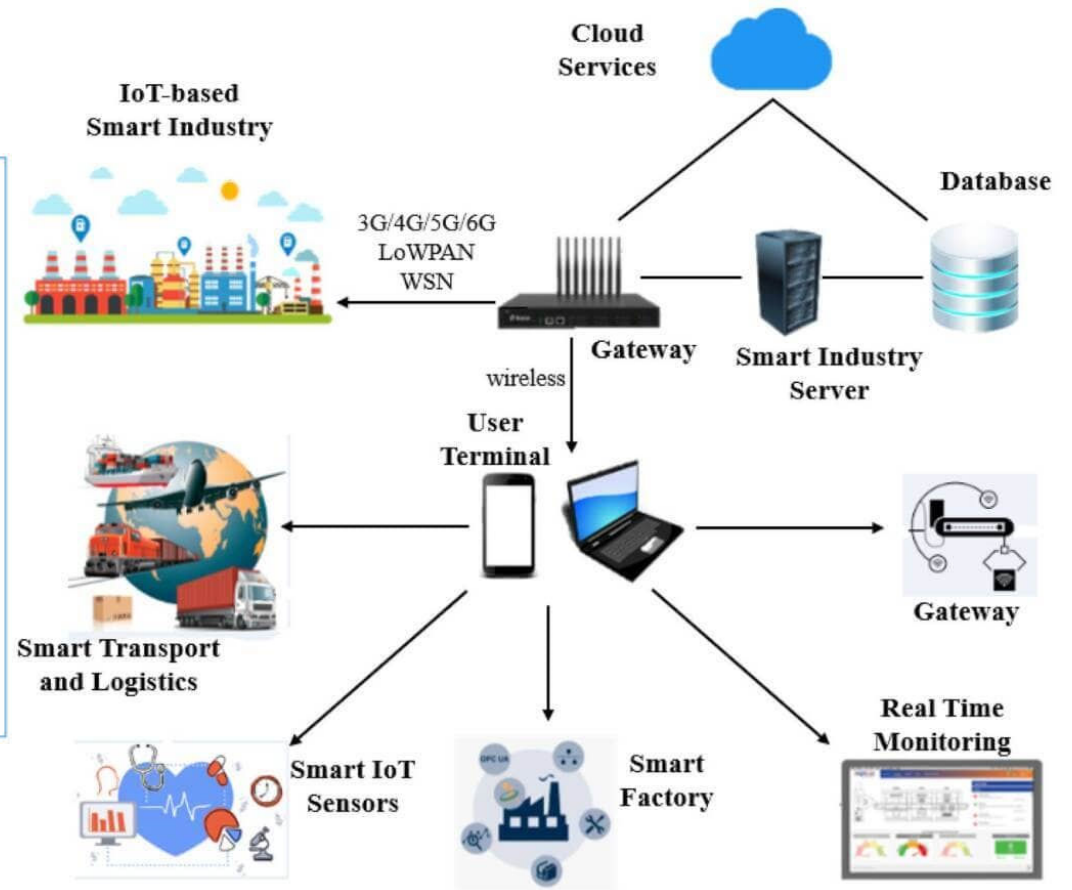
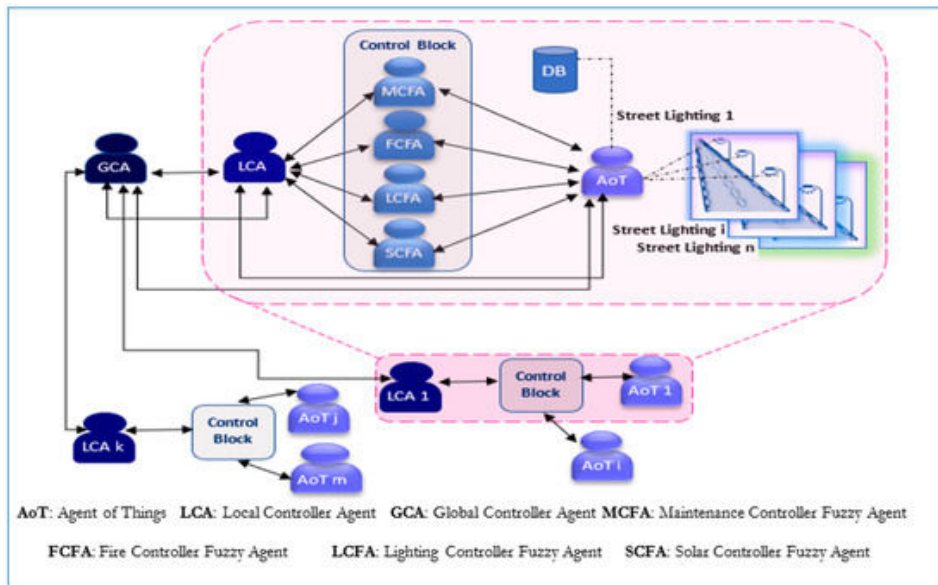
IoT and Intelligent Distributed Agent

Relationship Between IoT and Intelligent Distributed Agents	
IoT	Intelligent Distributed Agents
Collects data	Analyzes data
Connects devices	Makes decisions
Uses sensors	Uses AI algorithms
Sends information	Coordinates actions
Provides communication	Provides intelligence

Think of it this way:

- IoT provides the eyes, ears, and communication (sensors and networks).
- Intelligent Distributed Agents provide the brain and teamwork (AI decision-making and coordination).

IoT and Intelligent Distributed Agent



Machine Learning

Major Model of Machine Learning

Supervised Learning Supervised learning learns a mapping from labeled input data to output labels so it can accurately predict future outcomes.

Unsupervised Learning: Unsupervised learning uses unlabeled data to discover hidden patterns or group similar data. and let the model to figure out by itself.

Reinforcement Learning: It is trial and error like teaching a dog by giving food for good manner and etc.

Machine Learning Process or Pipeline

First, collect data from a real-world application or a random generator.

Next, clean the data by removing duplicate rows, outliers, and empty values, and apply feature engineering to improve it.

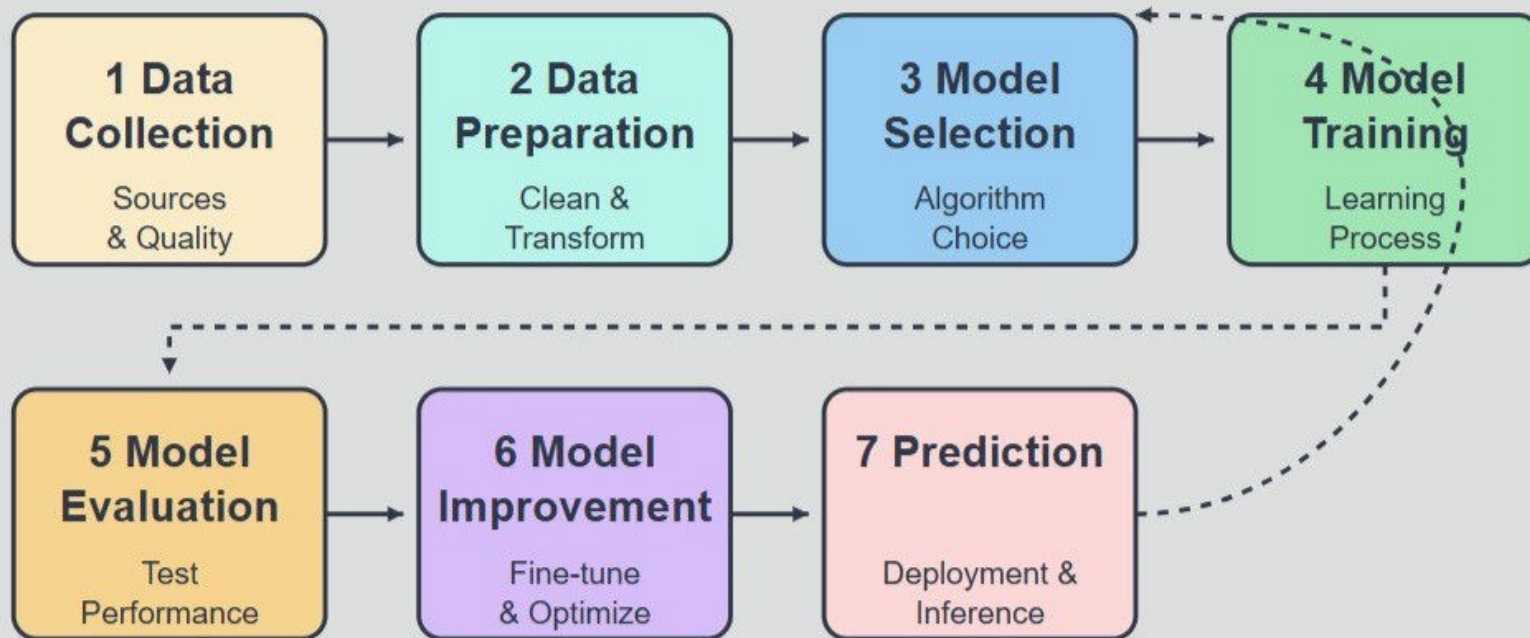
We use models such as Linear Regression, Clustering, and Logistic Regression to map input to output.

Let the model learn from data, that dim, and tune it to produce metrics such as accuracy.

We can use AutoML to automate all the above processes; an expert human must monitor for safety and ethical considerations.

Machine Learning

THE 7 STAGES OF MACHINE LEARNING A SIMPLE WORKFLOW



Machine Learning in Chip or Hardware

Machine Learning in chips or hardware refers to **designing computer hardware that can run machine learning algorithms efficiently**, often with lower power consumption and much higher speed than a general-purpose CPU.

CPUs are commonly used for data preprocessing, software development, and inference for smaller models.
and software development and inference.

Example AMD Ryzen 9 9950X and Intel Core Ultra 9

GPU is used for deep learning,, image recognition, large language model, and Video AI.

Example: NVIDIA GeForce RTX 5090

TPU is specialized processors optimized for TensorFlow and large-scale matrix operations.

Example: Google Search, Google Translate, Gemini, and Image Recognition

FPGA is used in Medical Equipment, Radar, Industrial Robots, and Network Devices

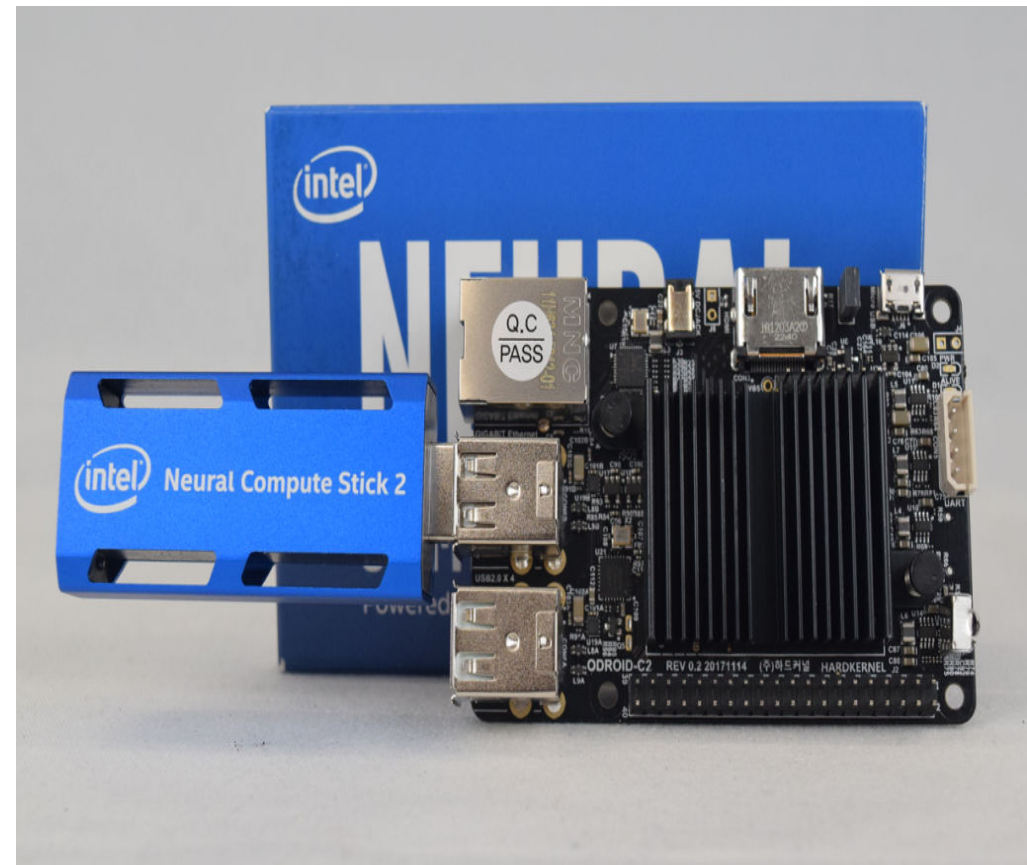
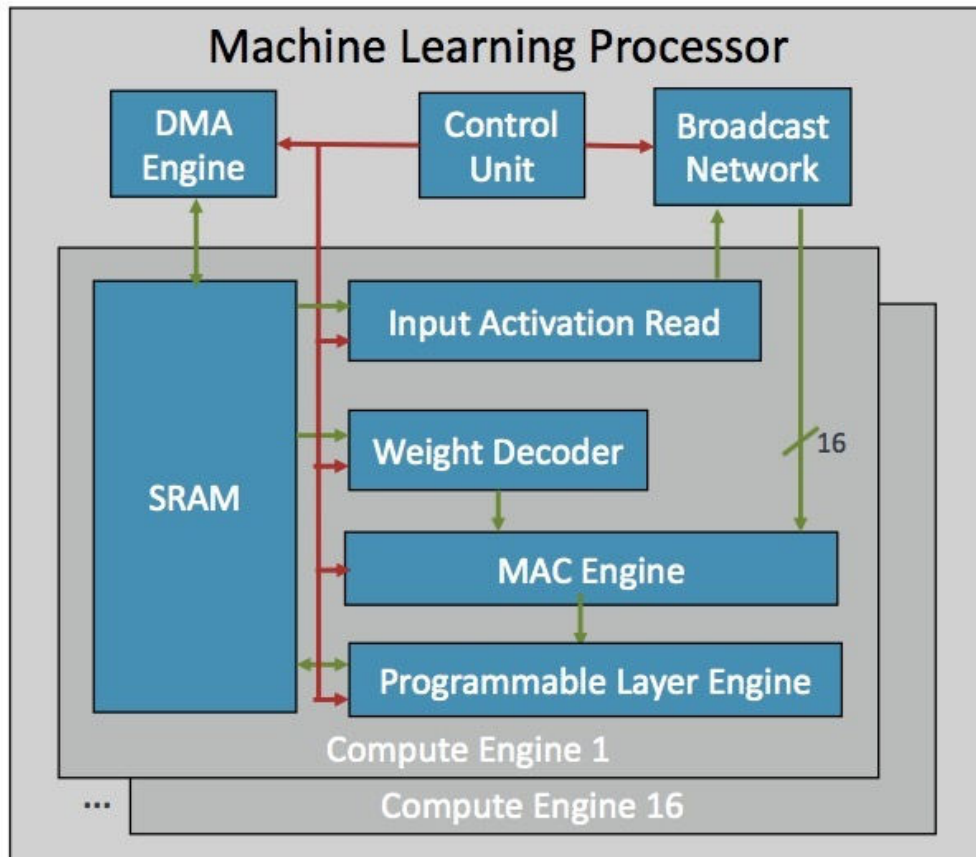
ASIC is used in Extremely Fast and Very Energy Efficient

Example: Smartphone AI processor, Bitcoin Mining Chip, and AI accelerator for Data Centers.

AI Hardware Architecture Process.

Sensor, AI Chip, Neural Network, and Prediction are process or flowchart of AI Hardware Design.

Machine Learning in Chip or Hardware



Neural Network

Neural Network

A **Neural Network (NN)** is a machine learning model inspired by the way the human brain processes information. It consists of interconnected processing units called neurons that work together to recognize patterns, make predictions, and solve complex problems.

Simple Definition

A neural network is a computer model that learns from examples by adjusting connections between artificial neurons to improve its predictions over time.

Comparing Traditional Machine Learning To Neural Network	
Traditional Machine Learning	Neural Network
Often requires manual feature engineering	Learns features automatically
Works well with smaller datasets	Usually performs better with larger datasets
Faster to train	Slower to train
Easier to interpret	Harder to interpret
Lower computational cost	Higher computational cost

Neural Network

Forward propagation in a Neural Network

